test statistic :

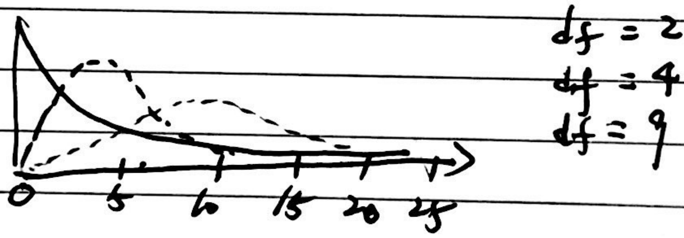
$$\frac{\text{point estimate} - \text{null value}}{\text{SE of point estimate}}$$

$$\boxed{\chi^2 \text{ statistic : } \chi^2 = \sum_{k=1}^{k} \frac{(O-E)^2}{E} \quad \begin{array}{l} O : \text{observed} \\ E : \text{expected} \\ k : \text{\# of cells} \end{array}}$$

- Why square?

  * positive standardized difference

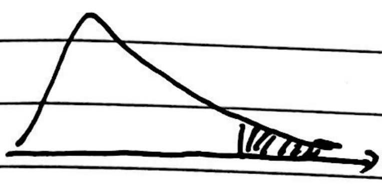
  * ampify unusual difference between expected and observed.



- p-value

  * p-value is defined as the tail area above the calculated test statistic

  * higher test statistic means higher deviation from null hypothesis.



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Chi-square tests

one categorical variable > 2 levels

① goodness of fit: (evaluating how well observed data fits expected)

A: compare distribution of <u>one</u> categorical variable (> 2 levels)
(compare to a hypothesized distribution)

B: degree of freedom: $df = k-1$ (k: # of cells)

② Independence:

A: evaluating the relationship between <u>two</u> categorical variables (at least one with more than two levels)

B: degree of freedom:
$$df = (R-1)(C-1)$$

* Conditions for the chi-square test:

	good of fit	Independence
① Independence		
* random sample/assign	✓	✓
* if sampling without replacement $n < 10\%$ of population	✓	✓
* each case only contributes to <u>one</u> cell in the table	✓	✓
② sample size/skew		
* each cell must have 5 expected cases	✓	✓

Chi-square test : Goodness of fit.

* Distribution of Ethnicities of people who are eligible for jury duty (based on census results)

ethnicity	white	black	nat. amer	asian & PI	other
% in population	80.29%	12.06%	0.79%	2.92%	3.94%

Distribution of 2500 people who are selected for jury duty in the previous year:

ethnicity	white	black	nat. amer.	asian & PI	other
observed #	1920	347	19	84	130

jury selection:

H_0: the observed counts of jurors from various race/ethnicities follow the same ethnicity distribution in the population (jurors are randomly selected)

H_A: Not randomly selected, counts do NOT follow the same race/ethnicity distribution.

ethnicity	white	black	nat. amer.	asian & PI	other	total
% in population	80.29%	12.06%	0.79%	2.92%	3.94%	100%
expect #	2007	302	20	73	98	2500

2500×0.8029 2500×0.1206

$2007 + 302 + 20 + 73 + 98 = 2500$ ✓

$$\chi^2 = \frac{(1920-2007)^2}{2007} + \frac{(347-302)^2}{302} + \frac{(19-20)^2}{20} + \frac{(84-73)^2}{73} + \frac{(130-98)^2}{98}$$

$= 22.63$

$df = k-1 = 5-1 = 4$

p-value $= 0.0002 \Rightarrow$ reject H_0

$\Rightarrow$ data provide convincing the observed distribution <u>do NOT</u> follow the distribution of population

Chi-Square Independent Test

* two categorical variable, at least one with > 2 levels

	dating	cohabiting	married	total
Obese	81	103	147	321
not obese	359	326	277	962
total	440	429	424	1293

Does there appear to be a relationship between weight and relationship status?

Ho: Weight and relationship status are independent.
Obesity rates do not vary by relationship status.

HA: Weight and relationship status are dependent.

χ^2 test of Independence $\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$

O_i : observed at i, E_i : expected at i
k : # of cells

$df = (R-1) \cdot \times (C-1)$ R: # of rows
C: # of colums

	col1	col2	...
Obese			
Not obese			

↑ two-way table

① What is the overall obesity rate in the sample?

$331/1293 = 0.256$

② If weight and relationship status are independent (if H_0 is true) how many dating people we expect to be obese? How many for cohabiting and married?

dating : $440 \times 0.256 \approx 113$

cohabiting : $429 \times 0.256 \approx 110$

married : $424 \times 0.256 \approx 108$

$113 + 110 + 108 = 331$ ✓

(making sure doing round properly)

⇒ (expected count)

expected count

	dating	cohabiting	married	total
obese	81 (113)	103 (110)	147 (108)	331
not obese	359 (327)	326 (319)	277 (316)	962
total	440	429	424	1293

Test the hypothesis that relationship status and obesity are associated at the 5% significance level

$$\chi^2 = \frac{(81-113)^2}{113} + \frac{(103-110)^2}{110} + \frac{(147-108)^2}{108} + \frac{(359-327)^2}{327}$$

$$+ \frac{(326-319)^2}{319} + \frac{(277-316)^2}{316} = 31.68$$

⇒ reject H_0 in favor of H_a

$df = (2-1)\times(3-1) = 2$, p-value $\doteq 1.3\times10^{-7}$ ⇒ data provided convincing evidence that relationship status and obesity are associated.